

## CHIBOUGAMAU CHAPAIS, QC, Canada

WMO: 718220

Lat: 49.7773N

Lon: 74.5303W

Elev: 387

StdP: 96.76

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |            |              | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|------------|--------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%         |              |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS         | MCDB         | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-34.7 | (c)<br>-31.9 | (d)<br>-39.6                  | (e)<br>0.1 | (f)<br>-34.5 | (g)<br>-36.7 | (h)<br>0.1 | (i)<br>-31.7 | (j)<br>9.1            | (k)<br>-9.3 | (l)<br>8.3 | (m)<br>-11.0 | (n)<br>1.2            | (o)<br>230 | (p)<br>0.695 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>10.8            | (c)<br>27.5     | (d)<br>18.3 | (e)<br>25.7 | (f)<br>17.3 | (g)<br>23.9 | (h)<br>16.5 | (i)<br>20.0         | (j)<br>24.5 | (k)<br>19.0 | (l)<br>23.1 | (m)<br>18.0 | (n)<br>21.9 | (o)<br>4.7           | (p)<br>220 |

| Dehumidification DP/MCDB and HR |             |             |             |             |             |             |             |             | Enthalpy/MCDB |             |             |             |             |             | Extreme<br>Max WB |
|---------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|-------------|-------------|-------------------|
| 0.4%                            |             |             | 1%          |             |             | 2%          |             |             | 0.4%          |             | 1%          |             | 2%          |             |                   |
| DP                              | HR          | MCDB        | DP          | HR          | MCDB        | DP          | HR          | MCDB        | Enth          | MCDB        | Enth        | MCDB        | Enth        | MCDB        |                   |
| (a)<br>18.6                     | (b)<br>14.1 | (c)<br>21.8 | (d)<br>17.6 | (e)<br>13.2 | (f)<br>20.9 | (g)<br>16.5 | (h)<br>12.3 | (i)<br>19.9 | (j)<br>59.3   | (k)<br>24.6 | (l)<br>55.8 | (m)<br>23.0 | (n)<br>52.5 | (o)<br>21.9 | (p)<br>25.0       |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 8.5               | 7.6  | 6.9 | -39.3                      | 30.8 | 2.4                | 1.7 | -41.0                                              | 32.0 | -42.5      | 33.0 | -43.8      | 34.0 | -45.6      | 35.3 |
|                   |      |     | -39.3                      | 22.0 | 2.4                | 1.2 | -41.0                                              | 22.9 | -42.5      | 23.7 | -43.8      | 24.3 | -45.5      | 25.2 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 0.6    | -18.0 | -16.5 | -9.6  | -1.2  | 7.9   | 14.0  | 16.5  | 15.2  | 10.8  | 3.9   | -4.2  | -12.6 |
|                                                                        | DBStd       | 13.43  | 7.95  | 7.65  | 7.65  | 5.70  | 5.31  | 4.57  | 3.24  | 3.51  | 4.55  | 4.65  | 5.67  | 7.27  |
|                                                                        | HDD10.0     | 4050   | 869   | 742   | 609   | 337   | 106   | 14    | 0     | 3     | 44    | 199   | 425   | 701   |
|                                                                        | HDD18.3     | 6523   | 1128  | 976   | 867   | 586   | 326   | 142   | 75    | 110   | 230   | 449   | 675   | 959   |
|                                                                        | CDD10.0     | 619    | 0     | 0     | 1     | 1     | 41    | 135   | 203   | 163   | 67    | 9     | 0     | 0     |
|                                                                        | CDD18.3     | 50     | 0     | 0     | 0     | 0     | 2     | 13    | 19    | 12    | 3     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 493    | 0     | 0     | 0     | 0     | 41    | 165   | 175   | 84    | 29    | 0     | 0     | 0     |
|                                                                        | CDH26.7     | 89     | 0     | 0     | 0     | 0     | 8     | 38    | 31    | 9     | 4     | 0     | 0     | 0     |
|                                                                        | WSAvg       | 3.3    | 3.1   | 3.3   | 3.5   | 3.6   | 3.5   | 3.3   | 3.2   | 3.1   | 3.3   | 3.6   | 3.5   | 3.1   |
| (15) Precipitation                                                     | PrecAvg     | 980    | 54    | 44    | 48    | 56    | 71    | 100   | 114   | 106   | 118   | 97    | 90    | 68    |
|                                                                        | PrecMax     | 1233   | 93    | 90    | 96    | 99    | 132   | 169   | 189   | 234   | 272   | 175   | 155   | 138   |
|                                                                        | PrecMin     | 749    | 13    | 0     | 0     | 12    | 19    | 34    | 42    | 38    | 59    | 37    | 35    | 2     |
|                                                                        | PrecStd     | 127    | 23    | 21    | 23    | 21    | 30    | 38    | 40    | 44    | 48    | 36    | 30    | 31    |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 4.0   | 4.4   | 10.9  | 16.0  | 27.4  | 30.4  | 29.8  | 28.0  | 26.7  | 20.1  | 10.2  | 3.1   |
|                                                                        |             | MCWB   | 3.7   | 2.8   | 6.0   | 7.8   | 16.8  | 19.0  | 19.6  | 19.4  | 19.6  | 14.3  | 7.8   | 2.4   |
|                                                                        | 2%          | DB     | 0.3   | 0.8   | 6.4   | 12.9  | 23.8  | 27.3  | 27.1  | 25.6  | 23.4  | 16.4  | 7.5   | 1.0   |
|                                                                        |             | MCWB   | 0.0   | -0.4  | 2.3   | 6.1   | 14.0  | 17.3  | 18.3  | 18.2  | 17.6  | 12.6  | 6.3   | 0.4   |
|                                                                        | 5%          | DB     | -3.2  | -1.3  | 3.5   | 9.8   | 20.8  | 25.1  | 25.3  | 23.7  | 20.8  | 13.5  | 5.0   | -0.3  |
|                                                                        |             | MCWB   | -3.9  | -2.4  | 0.8   | 4.5   | 12.6  | 16.3  | 17.2  | 17.3  | 16.1  | 11.1  | 3.9   | -0.8  |
|                                                                        | 10%         | DB     | -5.8  | -4.1  | 1.3   | 7.1   | 17.7  | 22.7  | 23.3  | 21.9  | 18.5  | 10.9  | 2.9   | -2.2  |
|                                                                        |             | MCWB   | -6.5  | -5.2  | -0.5  | 3.0   | 11.0  | 15.3  | 16.7  | 16.5  | 14.9  | 8.7   | 2.2   | -2.7  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 3.7   | 3.0   | 7.0   | 9.8   | 18.0  | 20.7  | 21.7  | 21.1  | 20.2  | 15.5  | 9.1   | 2.8   |
|                                                                        |             | MCDB   | 3.9   | 4.0   | 8.5   | 13.4  | 24.8  | 26.6  | 26.5  | 25.3  | 24.2  | 18.6  | 9.8   | 3.0   |
|                                                                        | 2%          | WB     | 0.0   | -0.2  | 3.0   | 6.8   | 15.3  | 19.2  | 20.0  | 19.6  | 18.6  | 13.2  | 6.3   | 0.5   |
|                                                                        |             | MCDB   | 0.3   | 0.7   | 5.3   | 10.7  | 20.4  | 24.3  | 24.3  | 23.3  | 22.1  | 15.7  | 7.1   | 0.9   |
|                                                                        | 5%          | WB     | -3.9  | -2.4  | 1.1   | 5.1   | 13.5  | 17.9  | 18.8  | 18.6  | 17.0  | 11.2  | 4.0   | -0.9  |
|                                                                        |             | MCDB   | -3.3  | -1.4  | 3.2   | 9.0   | 19.0  | 22.6  | 22.8  | 22.2  | 20.1  | 13.4  | 4.7   | -0.4  |
|                                                                        | 10%         | WB     | -6.5  | -5.1  | -0.4  | 3.5   | 11.8  | 16.5  | 17.7  | 17.4  | 15.1  | 9.1   | 2.2   | -2.8  |
|                                                                        |             | MCDB   | -5.7  | -4.2  | 1.4   | 6.6   | 17.3  | 21.0  | 21.7  | 20.7  | 18.1  | 10.8  | 2.9   | -2.2  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 10.9  | 12.5  | 12.6  | 11.0  | 12.2  | 12.0  | 10.8  | 10.1  | 9.8   | 7.1   | 6.5   | 8.7   |
|                                                                        | 5% DB       | MCDBR  | 10.9  | 10.6  | 12.9  | 14.6  | 16.9  | 15.2  | 13.8  | 12.8  | 12.7  | 11.4  | 8.4   | 7.3   |
|                                                                        |             | MCWBR  | 10.8  | 9.9   | 10.1  | 9.2   | 8.8   | 7.4   | 6.7   | 6.5   | 7.7   | 8.1   | 7.2   | 7.4   |
|                                                                        | 5% WB       | MCDBR  | 11.0  | 10.3  | 11.1  | 12.9  | 14.4  | 12.4  | 11.1  | 10.5  | 11.3  | 10.6  | 7.8   | 6.9   |
|                                                                        |             | MCWBR  | 11.0  | 9.8   | 9.1   | 9.0   | 8.6   | 7.0   | 6.4   | 6.3   | 7.6   | 8.1   | 7.3   | 7.0   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.251 | 0.266 | 0.279 | 0.327 | 0.337 | 0.357 | 0.372 | 0.355 | 0.347 | 0.305 | 0.285 | 0.244 |
|                                                                        | taud        |        | 2.267 | 2.244 | 2.260 | 2.340 | 2.474 | 2.448 | 2.428 | 2.476 | 2.474 | 2.576 | 2.466 | 2.333 |
|                                                                        | Ebn at Noon |        | 805   | 882   | 934   | 918   | 922   | 901   | 881   | 882   | 852   | 841   | 768   | 768   |
|                                                                        | Edh at Noon |        | 69    | 92    | 109   | 113   | 104   | 108   | 108   | 99    | 89    | 68    | 59    | 56    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.22  | 2.29  | 3.79  | 4.81  | 5.23  | 5.62  | 5.11  | 4.19  | 3.17  | 1.78  | 0.89  | 0.73  |
|                                                                        | RadStd      |        | 0.15  | 0.29  | 0.35  | 0.37  | 0.48  | 0.43  | 0.49  | 0.29  | 0.30  | 0.21  | 0.11  | 0.11  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | +1.75  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page